

CSS 5202: DEEP LEARNING AND APPLICATIONS [3 1 0 4]

Introduction to Deep Learning & Architectures, Machine learning basics, Neural Networks basics, Feed Forward Neural Networks, Machine Learning Vs. Deep Learning, Representation Learning, Width Vs. Depth of Neural Networks. Activation Functions: Linear, Non-Linear, Sigmoid, Tanh, RELU, LRELU, ERELU, SoftMax. Regularization and Optimization for Deep Learning, Convolutional Neural Networks Architectural Overview, Layers, Filters, Parameter sharing, Regularization. Popular CNN Architectures: ResNet, AlexNet. Transfer Learning: Transfer learning Techniques, Variants of CNN: DenseNet, PixelNet. Sequence Modelling: Recurrent and Recursive Nets, Recurrent Neural Networks, Bidirectional RNNs. Encoder-decoder sequence to sequence architectures, BPTT for training RNN, Long Short-Term Memory Networks. Auto Encoders, Regularized Autoencoders, stochastic Encoders and Decoders, Contractive Encoders. Deep Generative Models, Deep Belief networks, Boltzmann Machines, Deep Boltzmann Machine, Generative Adversarial Networks. Recent Trends. SDL: Transformer models, Explainable AI, Multimodal deep learning.

Reference Books

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville, " Deep Learning", MIT Press, 2017.
2. Josh Patterson, Adam Gibson "Deep Learning: A Practitioner's Approach", O'Reilly Media, 2017
3. Umberto Michelucci "Applied Deep Learning. A Case-based Approach to Understanding Deep Neural Networks" Apress, 2018.
4. Kevin P. Murphy "Machine Learning: A Probabilistic Perspective", The MIT Press, 2012.
5. Ethem Alpaydin, "Introduction to Machine Learning", MIT Press, Prentice Hall of India, Third Edition 2014.
6. Giancarlo Zaccaro, Md. Rezaul Karim, Ahmed Menshawy "Deep Learning with TensorFlow: Explore neural networks with Python", Packt Publisher, 2017.
7. Antonio Gulli, Sujit Pal "Deep Learning with Keras", Packt Publishers, 2017.
8. Francois Chollet "Deep Learning with Python", Manning Publications, 2017.
9. Rothman, Denis. Transformers for Natural Language Processing: Build innovative deep neural network architectures for NLP with Python, PyTorch, TensorFlow, BERT, RoBERTa, and more. Packt Publishing Ltd, 2021.
10. Samek, Wojciech, Grégoire Montavon, Andrea Vedaldi, Lars Kai Hansen, and Klaus-Robert Müller, eds. Explainable AI: interpreting, explaining and visualizing deep learning. Vol. 11700. Springer Nature, 2019.
10. Multimodal Deep Learning with Tensorflow, by Andrey But, Alexey Miasnikov, Gianluca Ortolani, Packt Publishing Limited, 2019.

Course Outcomes

At the end of the course, students must be able,

1. Differentiate the concept of machine learning with deep learning techniques for choosing suitable algorithms for different applications.
2. Understand the concept of CNN and transfer learning techniques to apply it in the classification problems.
3. Use RNN for time series prediction.

4. Use autoencoder and deep generative models to solve problems with high dimensional data including text, image, and speech.
5. Design and implement various machine learning algorithms in a range of real-world applications.